

Project - Expectation recorder

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(1) Understanding the Basics

(i) What is probability?

→ probability is the Branch of mathematics that measures the chance or likelihood of an event occurring. It is represented by a value between 0 and 1, where:

• 0 = impossible event

• 1 = certain event

Formula

$$P(E) = \frac{\text{Number of Favorable outcomes}}{\text{Total Number of Possible outcomes}}$$

(ii) Types of key probability Terminology

Term	Meaning
Experiment	An action that produces outcomes.
outcome	A possible result of an experiment.
sample spaces	The set of all possible outcomes.
Event (E)	A specific outcome or group of outcomes.
probability $P(E)$	The likelihood that an event occurs.
certain Event	An event that always happens ($P(E) = 1$).

impossible event

an event that never happens ($P=0$)

independent event

Event that do not each other

Dependent event

event where one affects that prob of another. bility

(C) Three probability event Examples from the Dataset.

Assume the dataset contains the following fields.

- study hours
- Attendance
- Group Discussion
- previous Test score
- Final Exam Pass.

Ex 1

Event: A randomly selected student has passed the final exam.

$$P(\text{Pass}) = \frac{\text{Number of Stud who pass}}{\text{Total number of student}}$$

Ex 2

Event: A randomly selected student has attendance greater than 80%

$P(\text{Attendance} > 80\%) = \frac{\text{students with attendance} > 80\%}{\text{Total students}}$

Ex 3

Event: A randomly selected student studies more than 5 hours per day

$P(\text{Study Hours} > 5) = \frac{\text{students studying} > 5 \text{ hours}}{\text{Total Students}}$

(2) Types of Events

(a) Empirical probability

Definition:

Empirical probability is based on actual data

Ex

in a dataset of 200 students, 150 students passed.

$$P(\text{Pass}) = \frac{150}{200} = 0.75$$

Answer: Probability of passing = 0.75 (75%)

(b) Theoretical probability

Definition:

Theoretical probability is Based on equal likely outcomes.

Ex

Rolling a fair die.

Event: Getting an even number.

$$P(\text{Even}) = \frac{3}{6} = 0.5$$

Answer: - probability of getting an even number = 0.5 (50%)

Difference.

- Empirical probability: - Based on actual data.
- Theoretical probability: - Based on mathematical rules.

(3) Random variable & probability Distribution.

(a) Define a Random Variable

Let x = Number of students who pass the final exam out of 3 randomly selected students.

possible values x are:

$$x = 0, 1, 2, 3$$

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(b) probability distribution Table

x students passed	probability P(x)
0	0.02
1	0.04 0.14
2	0.42
3	0.42

(These values are based on the pass probability $P = 0.75$.)

(c) Mean of variance

using the formula for a Binomial Distribution

- Number of students (n) = 3
- probability of passing (P) = 0.75

Mean

$$\text{Mean} = np = 3 \times 0.75 = 2.25$$

variance

$$\begin{aligned}\text{variance} &= np(1-p) \\ &= 3 \times 0.75 \times 0.25 = 0.5625\end{aligned}$$

Final answer

Random variable: x = Number of students passing out of 3 students.

$$\text{Mean} = 2.25$$

$$\text{variance} = 0.5625$$

(4) Venn diagram in probability (d)

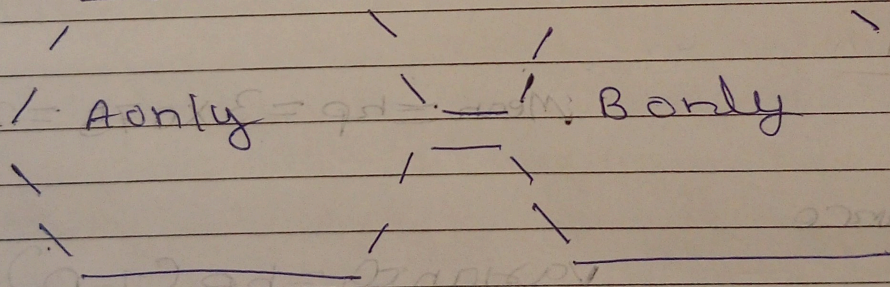
Let:

- A = students who study more than 10 hours/week.
- B = students who attend more than 80% of classes.
- A only: students who study more than 10 hours only.
- B only: students who attend more than 80% only.
- $A \cap B$: students who study more than 10 hours and attend more than 80%.

Simple Venn Diagram:

(Study ≥ 10 hrs)

(Attendance $\geq 80\%$)



$A \cap B$ = students satisfying both conditions

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(5) contingency Table & probability calculation

(a) contingency Table

Group Discussion	pass	Fail	Total
Yes	90	20	110
No	60	30	90
Total	150	50	200

(b) Joint probability.

Definition: probability that a student participates in group discussion and passes.

$$P(\text{Yes and pass}) = \frac{90}{200} = 0.45$$

Answer: 0.45 (45%)

(c) Marginal probability

Definition: - probability that a student passes the exam.

$$P(\text{Pass}) = \frac{150}{200} = 0.75$$

Answer: 0.75 (75%)

(d) Conditional probability.

Definition: probability of passing when the student participated in group discussion

$$P(\text{Pass} | \text{Yes}) = \frac{90}{110} = 0.82$$

Answer: 0.82 (82%)

(6) Understanding Relationships

(a) Interpretation of conditional probability.

Meaning -

Conditional probability tells us the chance of passing the exam when a student participates in a group discussion.

From the table

$$P(\text{Pass} | \text{Group Discussion}) = \frac{90}{110} = 0.82$$

Answer

A student who participated in a group discussion has an 82% chance of passing the exam.

(b) Relationship Between Events

The events "participating in group discussion" and "passing the exam" are dependent events.

Reason

participating in a group discussion changes the probability of ~~passing~~

passing the exam.

Answer:

Dependent Events (NOT independent and not mutually Exclusive)

(7) Bayes' Theorem Application.

- $P(\text{High Attendance} | \text{Pass}) = 0.70$
- $P(\text{High Attendance} | \text{Fail}) = 0.40$
- $P(\text{High Attendance}) = 0.60$
- Assume $P(\text{Pass}) = 0.75$ (from the dataset)

Formula:

$$P(\text{Pass} | \text{High Attendance}) = \frac{P(\text{High Attendance} | \text{Pass}) \times P(\text{Pass})}{P(\text{High Attendance})}$$

~~get~~

calculation:

$$= \frac{0.70 \times 0.75}{0.60} = \frac{0.525}{0.60} = 0.875$$

Answer:

$$P(\text{Pass} | \text{High Attendance}) = 0.875 = 87.5\%$$

conclusion:

A student with high attendance ($\rightarrow 80\%$) has an 87.5% chance of passing the exam.